

AUTOMATIC RAILWAY GATE CONTROL SYSTEM USING IR SENSOR

PROJECT EXPO 2026 – TEAM 2

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ABSTRACT

This project presents an automatic railway gate control system using two IR sensor modules, an Arduino Uno, and an SG90 servo motor. The sensors detect the arrival and departure of a train, while the Arduino controls the servo motor to close and open the railway gate automatically. A breadboard and jumper wires are used for the prototype circuit connections. The system demonstrates the basic concept of automatic gate operation and can be developed further for real railway applications.

INTRODUCTION

Railway crossings require proper gate control to reduce the risk of accidents. Manual operation can depend on timely human action. This prototype uses IR sensors to detect a train and a microcontroller to operate the gate automatically. When a train approaches, the gate closes, and after the train passes, the gate opens again.

OBJECTIVES

- To detect an approaching train using IR sensors.
- To automatically close the railway gate when a train is detected.
- To reopen the gate after the train has passed.
- To demonstrate an automatic railway gate prototype using simple electronic components.

COMPONENTS USED

Component	Quantity	Purpose
Arduino Uno board	1	Main controller
IR sensor modules	2	Train detection
SG90 servo motor	1	Open/close the model gate
Breadboard	1	Prototype circuit assembly
Jumper wires	As needed	Electrical connections

WORKING PRINCIPLE

The railway gate remains open normally. IR Sensor 1 is placed to detect an approaching train. When the sensor detects the train, the Arduino receives the signal and commands the SG90 servo motor to close the gate. The gate remains closed while the train passes. IR Sensor 2 detects the departure of the train, after which the Arduino commands the servo motor to open the gate.

PIN CONNECTIONS

Device	Arduino Pin
IR Sensor 1	D2
IR Sensor 2	D3
Servo Motor	D9
Power	5V and GND

REAL-TIME REPLACEMENTS

Prototype Component	Real-Time Railway Replacement
2 IR Sensors	Railway-grade train detection system
SG90 Servo Motor	Heavy-duty geared motor / actuator
Arduino Uno	Railway signalling / control system

TESTING AND RESULT

The prototype was tested by placing an object in front of the first IR sensor to simulate an approaching train. The Arduino detected the signal and the servo motor moved the gate to the closed position. After the object passed the second sensor, the servo motor returned the gate to the open position. The prototype successfully demonstrated automatic gate control.

ADVANTAGES

- Automatic gate operation
- Simple control using IR sensors and Arduino
- Low-cost prototype
- Reduces dependence on manual gate operation

LIMITATIONS

The SG90 servo motor and IR sensors are suitable for a small prototype but are not intended for full-size railway barriers. Real railway deployment requires stronger actuators, railway-grade detection, reliable control systems, backup power, environmental protection, fault monitoring, safety validation, and approval.

FUTURE SCOPE

The prototype can be upgraded with railway-grade train detection, heavy-duty geared motors or actuators, redundant sensors, backup power, fault monitoring, weatherproof enclosures, and safety-rated railway signalling/control systems.

CONCLUSION

The Automatic Railway Gate Control System using IR Sensors demonstrates how a train can be detected and a railway gate can be operated automatically. The prototype uses an Arduino Uno, two IR sensors, an SG90 servo motor, one breadboard, and jumper wires as needed. The concept can serve as a foundation for a more robust railway-grade automatic gate system.